

Assignment No- 01

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Answer to the question No 1:

The central dogma of molecular biology describes the flow of genetic information within a biological system. Here the main polymers are deoxyribonucleic acid (DNA), ribonucleic acid (RNA) and proteins. The main 3 processes here involved are DNA replication, transcription, Translation.

DNA (Deoxyribonucleic Acid):

Monomers: Nucleotides

A nucleotide consists of three components: a phosphate group, a five-carbon sugar (deoxyribose), and a nitrogenous base (adenine, thymine, cytosine, or guanine).

DNA is a structure of two polynucleotide strands that are held together by hydrogen bonds between nitrogenous bases (A with T and C with G). The sugar-phosphate backbone forms covalent phosphodiester bonds, linking the nucleotides together.

RNA (Ribonucleic Acid):

Monomers: Nucleotides

Similar to DNA, a nucleotide of RNA consists of a phosphate group, a five-carbon sugar (ribose), and a nitrogenous base (adenine, uracil, cytosine, or guanine).

RNA is typically single-stranded and can fold upon itself due to hydrogen bonding between complementary regions of the molecule. Like DNA, RNA also has phosphodiester bonds linking the nucleotides in the backbone. RNA converts genetic information from DNA to Proteins.

Proteins:

Monomers: Amino acids

Amino acids consist of a central carbon atom bonded to an amino group ($-NH_2$), a carboxyl group ($-COOH$), a hydrogen atom, and a variable side chain (R group). There are 20 different amino acids, each with a unique side chain.

Proteins are formed through peptide bonds, which are covalent bonds formed between the amino group of one amino acid and the carboxyl group of another. These peptide bonds link amino acids together to form a linear polypeptide chain, which is a structure such as hydrogen bonding.

The bonds involved in these polymers significantly influence their structure and function. For instance, the specific sequence of nucleotides in DNA and RNA determines the genetic identity, which in turn determines the sequence of amino acids in proteins. The structure of proteins, dictated by the sequence of amino acids, ultimately determines their function within the cell and organism. Therefore, the bonds involved in these polymers play crucial roles in transmitting and genetic information and conducting functions of organisms.

Answer to the question No 2:

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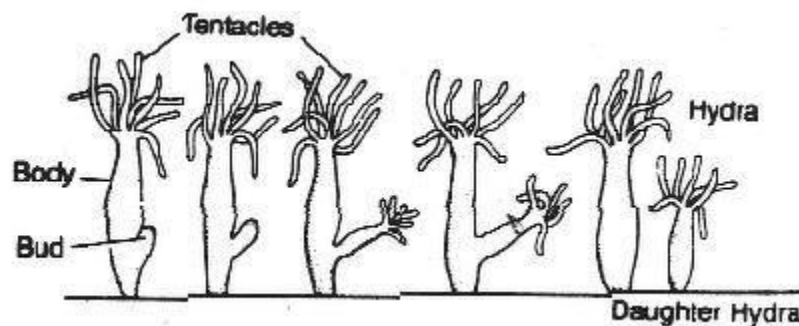
Food chain from the food-web, containing four organisms is Plankton -Krill-Herring-Seal in this food chain, the producer is plankton. Plankton are microscopic organisms that form the base of many marine food webs. Krill, Herring, and Seal are consumers in this food chain. Krill feed on plankton, making them primary consumers. Herring are secondary consumers that feed on krill, while Seal are tertiary consumers that feed on mackerel.

Food chain plays an important role in the eco system. They show how energy from the sun is captured by producers through photosynthesis and then transferred from one organism to another. They show how energy from the sun is captured by producers (such as plants or plankton) through photosynthesis and then transferred from one organism to another as it moves up the chain. Food chains help maintain balance and stability within ecosystems. They regulate population size and prevent any single species from becoming too dominant. When organisms consume other organisms, they break down organic matter into simpler compounds, which are eventually decomposed by bacteria and fungi. These decomposers release nutrients back into the environment, which can then be taken up by producers to restart the cycle. This nutrient cycling is vital for maintaining the health and productivity of ecosystems which is maintained by food chain.

Answer to the question no 3

Budding is an asexual procedure of producing new organisms. In this process, a new organism is developed from a small part of the parent's body. In hydra's body budding happens by Mitosis cell division. Mitosis is one of the process of asexual reproduction, where identical cells are produced from a single parent organism without the involvement of gametes. When the bud grows, the cells undergo rapid mitotic divisions to increase in number. This allows the bud to develop into a mature Hydra. Mitosis is essential for growth and development by providing a mechanism for increasing cell numbers while maintaining genetic identity. Mitosis ensures that each cell produced during cell division receives an exact copy of the parent cell's genetic material. As a result, the bud is genetically identical to the parent organism.

On the other hand, Meiosis cell division is production of gametes, such as sperm and egg cells, which is a part of sexual reproduction to form a zygote. Since Hydra reproduces asexually, there is no need for the formation of gametes, and thus no requirement for meiosis. Hydra's budding is a complex process which is a combination of cell differentiation, growing of new organs and Mitosis. But Meiosis cannot play any role in budding.



Reference

Brinkley, L. L. (1974). Non-Budding Hydra: Form Regulation and Bud Induction. *American Zoologist*, 14(2), 603–618. <http://www.jstor.org/stable/3882055>

